KUTAISI INTERNATIONAL UNIVERSITY

Scientific Computing

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#2.1

ODE Euler's method

- Convergence of explicit Euler's method
- Gronwall lemma, continuous case
- Gronwall lemma, discrete case
- Application of Gronwall lemma
- Convergence of implicit Euler's method
- Q & A

Theorem: Convergence of Explicit Euler's Method

Theorem 2.1 (Convergence of explicit Euler's Scheme)

- *Problem:*

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- *Convergence:* $|e_n| = O(h)$, $\Rightarrow$ *the method converges with order 1.*

Proof of the Convergence Theorem

Step 1: Definition of Local Truncation Error

- Let $u(t) \in C^2$ be the exact solution. $\Rightarrow$

$$u(t_{n+1}) = u(t_n) + hf(t_n, u(t_n)) + \frac{h^2}{2}u''(\xi_{n+1}), \quad \xi_{n+1} \in (t_n, t_{n+1}).$$

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- Subtract the Euler step from the Taylor expansion:

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- Taking norms: $\Rightarrow$

$$|e_{n+1}| \leq |e_n| + hL|e_n| + |\tau_{n+1}| = (1 + hL)|e_n| + |\tau_{n+1}|.$$

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$$|e_n| \leq \frac{Mh^2}{2} \sum_{k=0}^{n-1} e^{Lhk}.$$

Proof of the Convergence Theorem (Conclusion)

Step 5: Evaluating the Geometric Series

- The sum is a geometric series:

$$\sum_{k=0}^{n-1} e^{Lhk} = \frac{e^{Lhn} - 1}{e^{Lh} - 1}.$$

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- $e^x - 1 = 1 + x + x^2/2e^{(}\xi) - 1, \xi > 0 \Rightarrow e^{Lh} - 1 \geq Lh$:

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- Substituting these bounds:

$$|e_n| \leq \frac{Mh^2}{2} \cdot \frac{e^{L(t_n - \alpha)} - 1}{Lh}.$$

Proof of the Convergence Theorem (Final Result)

Step 6: Final Error Bound

- Simplifying the expression:

$$|e_n| \leq \frac{Mh}{2L}\left(e^{L(t_n-\alpha)}-1\right).$$

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- Simplifying the expression:

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- M and L are constants independent of h, $t_n \in [\alpha, \beta]$:

$$|e_n| \leq C \cdot h, \quad C = \frac{M}{2L}(e^{L(\beta-\alpha)}-1).$$

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Conclusion: The global truncation error is proportional to h. Thus, the explicit Euler method is **first-order convergent**.

Continuous Grönwall Lemma

Lemma 2.2

Let $u(t)$ be continuous on $[t_0, T]$ and satisfy

$$u(t) \leq \alpha + \int_{t_0}^{t} \beta u(s) ds,$$

where $\alpha, \beta \geq 0$ are constants.

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$$\frac{d}{dt} w(t) = \beta u(t)e^{-\beta t} - \beta v(t)e^{-\beta t} = \beta e^{-\beta t}(u(t) - v(t)).$$

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6. Multiply both sides by $e^{\beta t}$:
$$v(t) \leq \alpha(e^{\beta(t-t_0)} - 1).$$
7. Substitute back into $u(t) \leq \alpha + v(t)$:
$$u(t) \leq \alpha + \alpha(e^{\beta(t-t_0)} - 1) = \alpha e^{\beta(t-t_0)}.$$

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- **Define:** $v_n = \sum_{k=0}^{n-1} u_k$ (so $v_0 = 0$).
- **Recurrence:** $u_n \leq \alpha + \beta v_n$.
- **Sum the recurrence for u_k from $k = 0$ to $n - 1$:**

$$v_n = \sum_{k=0}^{n-1} u_k \leq n\alpha + \beta \sum_{k=0}^{n-1} v_k .$$

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- **Recurrence:** $u_n \leq \alpha + \beta v_n$.
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- **Final bound for u_n:**

$$u_n \leq \alpha + \beta v_n \leq \alpha + \beta\left(\frac{\alpha}{\beta}\left((1+\beta)^n - 1\right)\right) = \alpha(1+\beta)^n.$$

General Gronwall's Lemma: Continuous vs. Discrete,

Continuous Gronwall Lemma:

- If $u(t) \leq a(t) + \int_0^t b(s)u(s)\,ds$, then

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Application: Error analysis for numerical ODEs (e.g., Euler method).

Error Recurrence for Explicit Euler Method

Recall the error recurrence from the explicit Euler method:

$$|e_{n+1}| \leq (1 + hL)|e_n| + |\tau_{n+1}|,$$

where:

- $e_n = y_n - u(t_n)$ (global error),
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Goal: Bound $|e_n|$ using the Discrete Gronwall Lemma.

Applying the Discrete Gronwall Lemma

Rewrite the unfolded error bound:

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Identify terms for Gronwall's Lemma:

- Let $\alpha_k = (1 + hL)^k |\tau_{n-k}|$,
- But this doesn't directly fit. Instead, consider the recurrence:

$$|e_{n+1}| \leq |\tau_{n+1}| + (1 + hL)|e_n|.$$

This is of the form:

$$e_n \leq \sum_{k=0}^{n-1} \alpha_k + \beta \sum_{k=0}^{n-1} e_k,$$

where:

- $\alpha_k = |\tau_{k+1}|$,
- $\beta = (1 + hL)$.

Final Error Bound via Gronwall

Apply the Discrete Gronwall Lemma to:

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Global Error: For $t_n = nh$, the error grows exponentially in t_n:

$$|e_n| \leq Ch^p e^{Lt_n}$$

Implications for Numerical Stability

Key Observations:

- The error bound depends on:
  - Local truncation error $|\tau_{k+1}|$,
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  - Step size h,
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Comparison with Continuous Case: In the continuous Gronwall lemma, the error grows as e^{Lt}, similar to the discrete case but without the h-dependence in the exponent.

Summary: Discrete Gronwall in Numerical ODEs

Steps:

1. Start with the error recurrence:

$$|e_{n+1}| \leq |\tau_{n+1}| + (1 + hL)|e_n|.$$

2. Recognize it as a form suitable for the Discrete Gronwall Lemma.
3. Apply Gronwall to obtain:

$$|e_n| \leq \left(\sum_{k=0}^{n-1} |\tau_{k+1}| \right) (1 + hL)^n.$$

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Takeaway: The Discrete Gronwall Lemma provides a **sharp bound** on the global error, revealing the interplay between local errors, stability, and step size.

Theorem: Convergence of Implicit Euler's Method

Theorem 2.4 (Convergence of implicit Euler's Scheme)

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- *Convergence:* $|e_n| = O(h)$, $\Rightarrow$ *the method converges with order 1.*

Proof of the Convergence Theorem

Step 1: Definition of Local Truncation Error

- Let $u(t) \in C^2$ be the exact solution. $\Rightarrow$

$$u(t_{n+1}) = u(t_n) + hf(t_{n+1}, u(t_{n+1})) + \frac{h^2}{2} u''(\xi_{n+1}), \quad \xi_{n+1} \in (t_n, t_{n+1}).$$

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- Subtract the implicit Euler step from the Taylor expansion:

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- Since $hL < 1$ (for stability), we have:

$$|e_{n+1}| \leq \frac{1}{1 - hL}|e_n| + \frac{|\tau_{n+1}|}{1 - hL}.$$

Proof of the Convergence Theorem (cont.)

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$$|e_N| \leq C \max_{0\leq k\leq N}|\tau_{k+1}| \sum_{k=0}^{N} C^{n-k} \leq C\frac{C^N-1}{C-1} \max_{0\leq kleqN}|\tau_{k+1}|.$$

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- **Key observation:** For the error bound, we only need:

Proof of the Convergence Theorem (cont.)

Step 4: Final Error Bound Derivation

- From Frame 3, we have for $e_0 = 0$:

$$|e_N| \leq C \frac{C^N - 1}{C - 1} \cdot \max_{0 \leq k \leq N-1} |\tau_{k+1}|$$

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- Finally, absorbing the $O(h)$ term into the constant:

$$|e_N| \leq \frac{M}{2L} \left(e^{L(t_N - \alpha)} - 1 \right) h + O(h^2)$$

Proof of the Convergence Theorem (cont.)

Step 5: Conclusion

- We have derived the global error bound:

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- **Key points:**
  - The method is consistent (local truncation error $O(h^2)$)
  - The convergence is global and of order 1

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- This proves first-order convergence of the implicit Euler method.
- **Key points:**
  - The method is consistent (local truncation error $O(h^2)$)
  - The convergence is global and of order 1
- Thus, the implicit Euler method converges with order 1 to the exact solution as $h \to 0$.

Q & A